Neurodegeneration as Electromagnetic Flow Failure:
A Unified Structural Theory and Field-Structured Biology Analysis

Architecture, Coherence, and the Collapse of Biological EM Organization in Alzheimer's Disease, Parkinson's Disease, ALS, Multiple Sclerosis, Frontotemporal Dementia, Lewy Body Disease, and Vascular Cognitive Impairment

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Abstract

Neurodegeneration has been investigated for over a century through the lens of molecular pathology: the accumulation of misfolded proteins, the loss of specific neurotransmitters, and the failure of cellular metabolism. Despite enormous investment in this paradigm, disease-modifying therapies have remained elusive across the full spectrum of neurodegenerative conditions, from Alzheimer's disease and Parkinson's disease to amyotrophic lateral sclerosis, multiple sclerosis, frontotemporal dementia, Lewy body disease, and vascular cognitive impairment. This paper proposes that the explanatory and therapeutic inadequacy of the molecular-deficit model stems from a fundamental misidentification of the primary level of biological failure. Drawing upon the theoretical frameworks of **Unified Structural Theory** (**UST**) and **Field-Structured Biology** (**FSB**), we advance the position that neurodegeneration is, at its organizing core, a failure of biological architecture, and that this architectural failure produces obligate collapse of the coherent **electromagnetic (EM) fields** through which neuronal systems organize, communicate, and maintain identity. When cytoskeletal integrity is compromised, when myelin sheaths degrade, when extracellular matrix composition is disrupted, and when glymphatic maintenance fails, the structural substrates that guide and sustain directed electromagnetic flow dissolve. The chemical abnormalities catalogued by conventional neuropathology are consequences of this field failure, not its primary causes. This paper presents a systematic application of UST and FSB principles to the full range of major neurodegenerative conditions, mapping the architecture-specific mechanisms of EM flow failure in each, deriving a symptom-level interpretation grounded in field-state disruption, and identifying the therapeutic implications that follow when structural restoration, rather than molecular suppression, is placed at the center of intervention logic.

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1. Introduction

The dominant framework governing neurodegenerative disease research for the past four decades is organized around a molecular-deficit hypothesis: specific proteins misfold, specific transmitters are depleted, specific genes are expressed abnormally, and the resulting biochemical disruptions produce neuronal death and cognitive decline. This framework has generated extraordinary descriptive richness. The identification of amyloid-beta and tau as pathological hallmarks of Alzheimer's disease, the characterization of alpha-synuclein aggregation in Parkinson's disease and dementia with Lewy bodies, the discovery of TDP-43 and FUS inclusions in amyotrophic lateral sclerosis and frontotemporal dementia, the elucidation of myelin oligodendrocyte glycoprotein as an autoimmune target in multiple sclerosis, these represent genuine scientific achievements. Yet the translation of this molecular knowledge into disease-modifying therapy has been, by any honest accounting, catastrophically disappointing. The near-universal failure of amyloid-clearing agents to arrest cognitive decline in Alzheimer's disease trials, the limited symptomatic relief offered by dopamine replacement in Parkinson's disease, and the absence of any meaningful therapy for ALS collectively expose the explanatory limits of a model that treats molecular chemistry as the primary level of biological organization in the nervous system.

Unified Structural Theory (UST) proposes a corrective reorientation. UST holds that biological function is fundamentally a product of geometric and structural integrity. Function does not emerge from chemistry alone, it emerges from chemistry organized within and by a structural substrate that constrains, directs, and amplifies molecular interactions. When that structural substrate degrades, function ceases not merely because a particular molecule is absent, but because the architecture that gives molecular reactions their directional coherence, their temporal precision, and their spatial localization has collapsed. Chemistry without structure is noise. Structure without chemistry is inert. In the living nervous system, the inseparability of these two dimensions has been systematically underweighted by a field that measures molecules with great precision while measuring architecture only incidentally.

Field-Structured Biology (FSB) extends UST into the electromagnetic domain. FSB posits that living tissue does not merely conduct electricity as an incidental consequence of ion movement, it maintains coherent electromagnetic fields that function as primary organizers of biological order. These fields are not epiphenomenal. They guide ion flow, modulate molecular signaling, coordinate cellular identity across spatial scales, and sustain the informational coherence of tissue systems. The EM field is the medium through which the structural geometry of the neuron, its cytoskeletal lattice, its membrane organization, its extracellular matrix enclosure, is translated into physiological function. Disease, in the FSB framework, is not simply the loss of a molecular component. It is the loss of **field coherence**: the dissolution of the organized electromagnetic state that defines healthy tissue.

This theoretical perspective has a distinguished, if underappreciated, intellectual history. Robert O. Becker's experimental work on regenerative bioelectricity established that living tissues maintain DC electrical fields whose disruption correlates with pathological states, and whose restoration supports tissue repair (Becker & Selden, 1985). Herbert Fröhlich's theoretical physics of biological coherence demonstrated that metabolically active biomolecular assemblies can sustain coherent electromagnetic oscillations far beyond thermal equilibrium, providing a physical basis for long-range biological coordination (Fröhlich, 1968; Fröhlich, 1988). W. Ross Adey's experimental investigations of extremely low-frequency electromagnetic fields on calcium efflux from neural tissue revealed that biological systems respond to EM fields at intensities and frequencies entirely inconsistent with simple thermal or electrolytic models, indicating the existence of organized, resonance-sensitive field structures in living neural tissue (Adey, 1981; Adey, 1993). More recently, Stuart Hameroff and Roger Penrose have articulated a model in which neuronal microtubules function as quantum-coherent EM processors, cylindrical lattice polymers whose tubulin subunits maintain dipole states capable of sustaining vibrational information processing across a hierarchy of frequencies from hertz to terahertz (Hameroff & Penrose, 1996; Hameroff et al., 2026).

The thesis of this paper is stated without equivocation: neurodegeneration is not primarily a chemistry problem. It is an architecture problem with electromagnetic consequences. The misfolded proteins, depleted transmitters, and failed metabolic cycles that define neurodegenerative pathology are downstream effects of structural collapse, collapse that begins in the cytoskeleton, propagates through the membrane, degrades the extracellular matrix, and ultimately eliminates the physical scaffolding upon which coherent EM fields depend. To treat neurodegeneration without restoring this architecture is to repaint a building whose foundation has failed. The sections that follow develop this argument systematically, applying UST and FSB principles to the anatomy, pathology, symptomatology, and potential therapeutic reconfiguration of the full spectrum of major neurodegenerative conditions.

2. The Structural Basis of Neural Electromagnetic Flow

2.1 Cytoskeletal Architecture as Electromagnetic Waveguide

The neuron is not simply a chemical reactor enclosed in a lipid membrane. It is an architecturally sophisticated electromagnetic device, a structure whose geometry has been optimized over hundreds of millions of years to sustain, direct, and modulate the flow of charge with extraordinary precision. At the core of this EM organization is the **cytoskeleton**: the dynamic lattice of protein polymers, principally microtubules, actin filaments, and neurofilaments, that defines neuronal shape, organizes intracellular space, and provides the physical medium through which EM signals propagate and are transduced into biological action.

Microtubules occupy a particularly important position in the FSB account of neural EM organization. These cylindrical polymers, assembled from alpha- and beta-tubulin heterodimers arranged in a helical lattice 25 nanometers in diameter, possess well-documented piezoelectric properties: mechanical deformation of the tubulin lattice generates electrical polarization, and electrical fields induce conformational changes in tubulin subunits (Hameroff & Penrose, 1996; Tuszyński et al., 1995). The C-terminal tails of tubulin subunits carry a large negative charge, creating an electrostatic environment within and around the microtubule cylinder that is sensitive to ionic concentration and pH, two parameters that shift dramatically during neuronal activity. The interior of the microtubule contains a structured water core whose hydrogen-bonding network exhibits dielectric properties distinct from bulk cytoplasmic water, providing a medium capable of sustaining coherent dipole oscillations (Hameroff et al., 2026). In aggregate, the microtubule lattice constitutes a **piezoelectric EM waveguide**: a structure that couples mechanical, electrical, and chemical signals along geometrically defined pathways, converting the neuron's physical architecture into a coherent electromagnetic processing unit.

Actin networks complement microtubule function as ion-guiding scaffolds at the nanoscale periphery of the neuronal architecture. The dense actin cortex beneath the plasma membrane constrains the diffusion of membrane-associated proteins and ion channels, creating localized domains of ionic concentration and electrical potential that correspond to functional microcompartments. Actin's polymerization state, regulated by activity-dependent calcium signaling, dynamically reshapes these compartments in real time, meaning that the EM microenvironment of any given membrane domain is a direct function of actin structural organization. In dendritic spines, actin architecture determines synaptic receptor density, AMPA receptor lateral mobility, and the geometry of calcium microdomains, all of which are simultaneously structural and electromagnetic phenomena (Bhatt et al., 2009; Hotulainen & Hoogenraad, 2010).

Neurofilaments, the intermediate filaments specific to neurons, establish the charge gradients along the length of axons that sustain their electrochemical polarity. The highly phosphorylated sidearms of neurofilament heavy chain (NF-H) create a fixed negative charge density within the axoplasm that influences the radial distribution of cations, the caliber of the axon, and the velocity of conduction. Neurofilament organization is not incidental to axonal function: it is the structural determinant of the axon's EM identity. Disruption of neurofilament stoichiometry, as occurs in ALS and other neurodegenerative conditions, degrades this charge gradient and impairs the EM coherence of the long-range conducting fiber (Cleveland & Rothstein, 2001).

2.2 Membrane Geometry and Field Coherence

If the cytoskeleton constitutes the internal EM waveguide of the neuron, the plasma membrane constitutes its field interface, the dynamic boundary at which intracellular EM organization encounters and responds to extracellular field conditions. Membrane EM function is not reducible to the opening and closing of ion channels. It is organized at the mesoscale by **lipid raft** domains: ordered, cholesterol-rich membrane microdomains of 10 to 200 nanometers in diameter that concentrate signaling proteins, modulate membrane fluidity, and create discrete zones of altered electrical permittivity (Simons & Vaz, 2004).

These rafts function as localized EM domains, membrane regions in which the dielectric properties, surface charge density, and protein composition combine to produce distinctive electrical boundary conditions. The coherent activity of lipid raft-organized signaling cascades depends on the precise spatial organization of these domains; disruption of raft composition through oxidative modification of cholesterol or altered sphingomyelin metabolism, both documented in Alzheimer's disease, fragments these EM microdomains and degrades their organizational function (Hicks et al., 2012).

Myelin represents the most architecturally sophisticated EM structure in the mammalian nervous system. The myelin sheath, a multilamellar lipid membrane wrapped concentrically around the axon by oligodendrocytes in the central nervous system and Schwann cells in the peripheral nervous system, functions as a biological capacitor: its high electrical resistance and low capacitance allow axonal membrane voltage to propagate with minimal loss across the internodal segments, confining active ion exchange to the geometrically precise interruptions of the **nodes of Ranvier**. From the FSB perspective, this architecture is not merely a metabolic efficiency strategy, it is a **structural EM amplifier**. The node of Ranvier, flanked by the paranodal junctions and juxtaparanodal potassium channel clusters, constitutes a precisely engineered EM relay station at which the field geometry of the axon is regenerated with high fidelity. The integrity of this relay function depends entirely on the physical architecture of the myelin wrap, the paranodal loops, and the perinodal extracellular matrix, none of which can be reproduced by pharmacology alone once structural disruption has occurred (Rasband & Bhatt, 2007; Fawcett et al., 2019).

2.3 Extracellular Matrix as Field Medium

The extracellular matrix (ECM) of the brain, long regarded as mere structural scaffolding, is, in the FSB framework, an active electromagnetic medium: a charge-organized, hydrated polymer gel through which field signals propagate, ions diffuse along structural gradients, and the identities of individual neurons are stabilized by their molecular enclosures. **Proteoglycans**, particularly heparan sulfate proteoglycans (HSPGs) and chondroitin sulfate proteoglycans (CSPGs), contribute dense arrays of negatively charged sulfate and carboxylate groups to the ECM, creating organized charge gradients that guide cation movement and modulate the EM environment of neuronal surfaces (Kwok et al., 2011). These charge distributions are not static; they are dynamically regulated by the activity of matrix metalloproteinases and heparanases, making the ECM a responsive, state-dependent EM medium rather than a passive scaffold.

Perineuronal nets (PNNs), the condensed, lattice-like ECM structures that envelop the soma and proximal dendrites of specific neuronal populations, particularly fast-spiking parvalbumin-positive interneurons, represent the most architecturally organized EM enclosure in the nervous system. PNNs, which are composed principally of CSPGs, hyaluronan, tenascin-R, and link proteins, create a molecular cage around the neuron that stabilizes ionic microenvironments, restricts the lateral mobility of membrane proteins, and functions as a **field-stabilizing enclosure** (Fawcett et al., 2019; Sakata et al., 2025). The loss or degradation of PNNs, documented in Alzheimer's disease, schizophrenia, and aging, exposes the enclosed neurons to EM perturbation and disrupts their capacity to maintain stable firing rates and synaptic integration (Howell et al., 2015).

The **glymphatic system**, the perivascular CSF transport network driven by aquaporin-4 (AQP4) water channels on astrocytic endfeet, must be understood within the FSB framework as the brain's primary **EM maintenance infrastructure**. During slow-wave sleep, when noradrenergic tone falls, extracellular space expands by approximately 60 percent, and a pressure-driven convective flow of CSF along perivascular spaces clears the metabolic byproducts, including amyloid-beta and tau, that accumulate during waking neural activity (Xie et al., 2013; Hauglund et al., 2025). From the FSB perspective, this clearance function is not merely metabolic housekeeping: it is the physical restoration of the ECM's charge organization and compositional integrity, the nightly reconstitution of the EM field medium upon which the following day's neural coherence depends. Glymphatic failure is, therefore, not simply metabolic failure; it is the chronic deterioration of the structural medium through which field-organized biology operates.

3. Mechanisms of Structural Collapse and Electromagnetic Flow Failure

3.1 Tau Pathology as Cytoskeletal Dissolution

Tau is a microtubule-associated protein (MAP) whose primary function under physiological conditions is the stabilization of axonal microtubule arrays. In healthy neurons, tau binds cooperatively to the microtubule lattice, forming protective molecular envelopes that shield microtubules from severing enzymes such as katanin, preserve lattice integrity under mechanical stress, and sustain the EM waveguide function described in the preceding section (Triclin et al., 2021; de Forges et al., 2024). Tau's EM significance derives precisely from this stabilizing role: without tau, microtubule arrays in long axonal processes are dynamically unstable, subject to catastrophic depolymerization events that eliminate the structural substrate of EM conductance along the entire length of the fiber.

In the tauopathies, which include Alzheimer's disease, frontotemporal dementia with tau inclusions (FTLD-tau), progressive supranuclear palsy, and corticobasal degeneration, tau undergoes **hyperphosphorylation** at multiple epitopes by kinases including CDK5, GSK-3beta, and DYRK1A. Hyperphosphorylated tau exhibits reduced affinity for the microtubule lattice: it dissociates from the microtubule surface, disrupts the cooperative envelope structure, and renders the microtubule vulnerable to severing (de Forges et al., 2024; Guo et al., 2017). This is not a secondary consequence of tau pathology, it is its primary structural action. The microtubule loses its stabilizing partner and, with it, its capacity to function as a coherent EM waveguide. Charge propagation along the axon becomes irregular, action potential conduction velocity decreases, and the precise temporal relationships between firing neurons, the oscillatory coherence on which network computation depends, begin to degrade.

As hyperphosphorylated tau accumulates in the soma, it forms the **neurofibrillary tangles (NFTs)** whose staging by Braak and colleagues has provided the definitive anatomical framework for Alzheimer's disease progression (Braak & Braak, 1991). In the UST/FSB analysis, NFTs are not simply markers of neuronal stress, they are **EM dead zones**: densely packed, insoluble protein aggregates that occupy the cytoplasmic space previously organized by functional microtubule arrays. The tangle-filled neuron has lost the internal EM architecture that made it a functional computational unit. Its remaining chemistry may persist for a time, mitochondria may continue to produce ATP, ribosomes may continue to synthesize protein, but the structural organization that gave those chemical processes their coordinated biological meaning has been eliminated.

3.2 Amyloid as Field Disruptor

The amyloid cascade hypothesis, first articulated by Hardy and Higgins (1992) and subsequently refined by Selkoe and Hardy (2016), assigns beta-amyloid (Abeta) a causal role at the apex of the Alzheimer's disease pathological hierarchy. Within the molecular-deficit framework, this hypothesis has motivated the most heavily funded therapeutic program in the history of neurological drug development, and the most documented series of clinical failures. The FSB framework offers an alternative interpretation that resolves several paradoxes in the amyloid literature while preserving its descriptive accuracy.

Beta-amyloid oligomers, the soluble, pre-plaque forms of Abeta now recognized as the most synaptotoxic species, disrupt membrane architecture through several mechanisms: they insert into lipid bilayers, alter membrane curvature, disrupt lipid raft organization, and impair the function of membrane-associated proteins including ion channels, receptors, and ECM-binding integrins (Lauren et al., 2009; Bhowmik et al., 2015). In FSB terms, oligomeric Abeta is a **membrane-geometry disruptor**: it degrades the precise spatial organization of the membrane EM interface, fragmenting the localized EM domains that support coherent synaptic signaling. This disruption precedes neuronal death by years, explaining why cognitive decline begins well before the histological criteria for neuronal loss are met.

The mature amyloid plaque represents a further level of ECM field disruption. Dense-core plaques embedded in the neuropil create zones of altered ECM composition, mechanical stiffness, and ionic concentration that distort the EM field medium in their immediate vicinity. Local neuronal processes, dendrites and axonal terminals, that encounter plaques must navigate an EM environment that is both compositionally abnormal and geometrically disrupted, producing the dystrophic neuritic changes and synaptic retraction that pathologists have documented for over a century.

Most critically, the FSB framework reframes the amyloid cascade itself: rather than treating Abeta accumulation as the initiating cause of Alzheimer's disease, it identifies amyloid deposition as a **symptom of prior field failure**. Specifically, the production and clearance of Abeta are themselves functions of cellular structural integrity, of the endosomal sorting pathways that depend on cytoskeletal organization, and of the glymphatic clearance that depends on ECM structural integrity and sleep architecture. When the underlying structural-field organization degrades, Abeta production-clearance balance fails, and amyloid accumulates. The plaque is the geological record of a structural catastrophe that preceded it, not its cause.

3.3 Protein Aggregation as Architectural Debris

The aggregation of misfolded proteins, **alpha-synuclein** in Parkinson's disease and dementia with Lewy bodies, **TDP-43** and **FUS** in ALS and frontotemporal dementia, represents, in UST terms, a class of structural load events: the accumulation of insoluble material within the cytoplasmic space that displaces functional architecture, imposes mechanical stress on the cytoskeletal lattice, and progressively eliminates the organized interior through which EM signals are guided. Alpha-synuclein, under physiological conditions, is a membrane-associated protein that facilitates synaptic vesicle trafficking, a function that requires precise interaction with lipid membrane geometry. In its aggregated, fibrillar form, it becomes a structural toxin: sequestering membrane lipids, disrupting mitochondrial architecture, and filling the synaptic terminal with insoluble debris that eliminates the structural prerequisite for synaptic EM function (Spillantini et al., 1997; Volpicelli-Daley et al., 2014).

TDP-43 and FUS are RNA-binding proteins whose nuclear and cytoplasmic distribution regulates mRNA processing, including the splicing of transcripts encoding cytoskeletal proteins. Their aggregation therefore constitutes a dual structural attack: as cytoplasmic aggregates, they impose physical load on the neuronal interior; as nuclear depletants, they deregulate the production of the very structural proteins, neurofilaments, tau isoforms, actin-regulatory proteins, upon which EM architecture depends (Cleveland & Rothstein, 2001; Lagier-Tourenne et al., 2010). The spread of protein aggregation between neurons, whether through exosomal secretion, synaptic transfer, or tunneling nanotubes, propagates structural load from cell to cell in a pattern that the FSB framework describes as **field collapse propagation**: the progressive elimination of architectural integrity across connected networks.

3.4 Myelin Degradation and Electromagnetic Insulation Failure

The demyelinating process in multiple sclerosis, and the subtler myelin degeneration that accompanies aging and other neurodegenerative conditions, constitutes the clearest and most anatomically explicit example of EM insulation failure in the nervous system. The myelin sheath functions as a biological faraday cage for the internode: it confines the axonal electrical field, prevents capacitative dissipation of signal energy across the internodal membrane, and delivers the concentrated EM pulse to the node of Ranvier with the precision required for temporally coherent network signaling. When myelin is destroyed, whether by autoimmune attack, as in MS, or by oligodendrocyte loss associated with aging, ischemia, or protein-aggregation toxicity, the internodal axon membrane becomes electrically exposed, and the insulated EM conduit becomes a leaky wire.

The consequences of **myelin insulation failure** are not merely quantitative reductions in conduction velocity. They represent qualitative changes in the EM structure of axonal signaling: the signal loses its temporal precision, its amplitude collapses with distance, and the precise phase relationships between firing neurons in large-scale networks are disrupted. Because network computation in the brain depends on coherent oscillations across spatially distributed populations of neurons, oscillations whose temporal precision is determined in large part by the conduction delays of myelinated fibers, myelin loss degrades the computational architecture of the brain at the level of its most fundamental EM circuit elements. Remyelination failure, which is the norm rather than the exception in progressive MS and in age-related white matter degeneration, constitutes the **irreversible architectural gap** that separates relapsing-remitting from progressive disease: once the structural basis of EM insulation is permanently absent, no pharmacological agent can restore the function it supported.

3.5 Vascular and Glymphatic Collapse

The structural integrity of the cerebral vascular system, particularly the blood-brain barrier (BBB), is a precondition for the maintenance of the ECM EM field medium. The BBB is not simply a permeability filter; it is a **structurally organized ionic boundary condition** that maintains the precise ionic composition of the brain interstitium, the medium through which EM fields propagate and neuronal surfaces are bathed. Tight junctions between cerebral endothelial cells, supported by pericyte structural investment and astrocytic endfeet, create a molecular architecture whose integrity is the prerequisite for ionic homeostasis in the neuropil. When this structural architecture degrades, as occurs in aging, hypertension, diabetes, and directly in vascular dementia, plasma proteins, inflammatory mediators, and ionic imbalances penetrate the brain parenchyma, constituting what the FSB framework characterizes as **field medium contamination**: the introduction of compositionally inappropriate material into the EM field environment of the neural tissue (Iadecola, 2017).

Glymphatic dysfunction compounds BBB failure by eliminating the restorative function that normally counteracts the structural degradation of the ECM field medium during waking hours. As glymphatic clearance efficiency declines, driven by AQP4 mislocalization, venous pulsatility reduction, and sleep architecture fragmentation, the ECM accumulates the metabolic detritus of neural activity: aggregation-prone peptides, oxidized lipids, inflammatory cytokines, and electrolytic waste. This accumulation progressively degrades the charge organization of the ECM, increases its effective electrical impedance, reduces the fidelity with which EM fields propagate through it, and creates the conditions under which protein aggregation is thermodynamically favored. In the FSB model, chronic glymphatic failure is therefore not simply a consequence of neurodegeneration, it is an accelerating structural cause of EM field medium deterioration that drives the pathological cascade forward.

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4. Field-State Disruption: From Local to Systemic

4.1 The Field-State Concept in Field-Structured Biology

The organizing concept of FSB at the tissue level is the **field-state**: a coherent electromagnetic organizational condition maintained across spatial scales by the integrated structural integrity of the cytoskeleton, membrane geometry, and ECM architecture of a neural population. A field-state is not a single measurable quantity, it is a distributed property of biological order, analogous to the ordered phase of a physical system. It encompasses the spatial distribution of ionic concentrations, the temporal pattern of membrane potential oscillations, the coherence of EM field fluctuations across spatially separated neurons, and the structural boundary conditions that sustain all of these. A healthy neural field-state is maintained continuously by the structural integrity of the tissue components described in the preceding sections. It supports synchronized oscillations, coherent information encoding, reliable synaptic transmission, and the maintenance of cellular identity across time.

The crucial distinction between the field-state concept and conventional molecular-state descriptions is one of causal priority. Molecular states, the phosphorylation state of a kinase, the binding state of a receptor, the glycosylation pattern of an ECM proteoglycan, are always embedded within and constrained by a field-state. The field-state is the higher-order physical context that determines which molecular transitions are possible, at what rates, and with what spatial coherence. Molecular-state descriptions that ignore this context inevitably generate incomplete causal models, models that can accurately describe what happens at a specific molecule in a specific cell but cannot account for the coordinated, network-level failures that define clinical neurodegeneration. In UST terms, structure precedes chemistry not only phylogenetically but causally: the structural-field context determines the chemical outcomes, not the reverse.

4.2 Local Field Collapse and the Spreading Lesion

Field collapse in the nervous system does not begin everywhere simultaneously. It initiates in anatomically specific, structurally vulnerable zones, regions whose architectural properties (high synaptic density, thin axonal caliber, high metabolic demand, complex dendritic geometry, or sparse myelination) make them disproportionately susceptible to the structural perturbations described above. From these initial foci, field collapse propagates outward along anatomically connected pathways in a manner that the FSB framework characterizes as **field-state contagion**: the progressive disruption of EM coherence in neurons that are structurally connected to regions of prior collapse.

This propagation mechanism is consistent with, and provides a biophysical explanation for, the prion-like spread of pathological protein conformers that has been documented across neurodegenerative conditions. In the conventional molecular account, misfolded alpha-synuclein, tau, or TDP-43 acts as a template that induces conformational changes in the normally folded protein of recipient cells, a seeding mechanism analogous to prion propagation. The FSB account adds an essential architectural dimension: the spread of misfolded protein conformers is simultaneously the spread of structural load events that progressively eliminate field coherence along anatomically connected pathways. The protein spread and the field-state collapse are not parallel processes, they are aspects of a single architectural catastrophe propagating through the structural network of the brain.

4.3 Network-Level Field Failure

The ultimate expression of field collapse in the neurodegenerative brain is the failure of large-scale EM networks, the loss of coordinated, oscillatory field coherence across the distributed neural populations that constitute the functional systems of cognition, movement, language, and behavior. The **default mode network (DMN)**, the collection of medial prefrontal, posterior cingulate, lateral parietal, and medial temporal regions that sustains self-referential cognition and autobiographical memory, is, in the FSB framework, a coherent EM circuit: a distributed population of neurons whose structural connectivity, myelination pattern, and oscillatory properties maintain a characteristic pattern of synchronized activity. In Alzheimer's disease, the DMN is the first large-scale network to exhibit measurable field-state degradation, a finding corroborated by the consistent observation of reduced functional connectivity within the DMN as the earliest network-level signature of the disease, detectable years before clinical symptom onset (Greicius et al., 2004; Jack et al., 2013).

The degradation of oscillatory coherence across the frequency bands of EEG, the progressive loss of gamma-band (30-80 Hz) coherence, the slowing and fragmentation of alpha rhythms (8-12 Hz), and the disruption of theta oscillations (4-8 Hz) in the hippocampal-entorhinal circuit, constitute the measurable, time-resolved signature of field-state collapse in the degenerating brain. These oscillatory changes are not secondary reflections of underlying molecular pathology, they are direct readouts of the EM organizational state of neural tissue, and they provide the most sensitive, temporally precise, and spatially informative biomarkers of neurodegenerative progression available to current clinical practice. The remarkable efficacy of 40 Hz gamma entrainment in amyloid clearance in preclinical Alzheimer's models (Iaccarino et al., 2016) supports the FSB interpretation: it demonstrates that externally imposed EM field organization can partially rescue biological function that structural collapse has begun to eliminate, confirming that EM field coherence is causally upstream of the molecular changes it organizes.

5. Architectural Breakdown by Region and Disease

5.1 Hippocampal Architecture and Memory: Alzheimer's Disease

The hippocampal formation and its primary afferent, the entorhinal cortex, exhibit an architectural vulnerability profile that explains their consistent priority of failure in Alzheimer's disease, a priority that Braak staging has documented with remarkable anatomical precision across thousands of post-mortem specimens. The entorhinal cortex, particularly layer II, whose stellate neurons project to the dentate gyrus via the perforant path, functions as the primary EM gateway between neocortical association areas and the hippocampal circuit. These neurons are characterized by long, thin, sparsely myelinated axons that project across considerable distances, high metabolic rates, dense dendritic trees, and limited perineuronal net investment, a combination of architectural properties that, in the FSB framework, constitutes a vulnerability signature: high EM throughput demand in a geometrically exposed structural context.

When tau hyperphosphorylation begins to compromise microtubule stability in entorhinal layer II neurons, Braak stage I, the perforant path begins to lose EM conductance integrity. The precisely timed theta-frequency (4-8 Hz) oscillations that the entorhinal-hippocampal loop uses to encode episodic memories become temporally imprecise, their phase relationships degraded by the slowed, variable conduction that characterizes a structurally compromised axon population. Memory encoding does not require only the presence of appropriate molecular signals, it requires the coherent, precisely timed EM coordination of entorhinal input with hippocampal CA3 recurrent collaterals and CA1 output neurons. When the EM timing of this circuit degrades, new memory traces cannot be formed with adequate fidelity, not because synaptic molecules are absent, but because the structural-field architecture that organizes their temporal sequence has deteriorated. The phenomenology of early Alzheimer's disease, the selective impairment of recent episodic memory with relative preservation of remote memories and procedural skills, maps precisely onto the anatomical progression of structural-field failure through the hippocampal circuit.

5.2 Substantia Nigra and Basal Ganglia Circuits: Parkinson's Disease

The dopaminergic neurons of the substantia nigra pars compacta (SNpc) possess an architectural vulnerability profile that is among the most pronounced in the central nervous system. Their axons are extraordinarily long and extensively branched, a single human SNpc dopaminergic neuron may maintain a total axonal arbor of several meters, with synaptic terminals numbering in the hundreds of thousands within the striatum (Bolam & Pissadaki, 2012). These axons are largely unmyelinated or thinly myelinated, and the metabolic cost of maintaining their EM architecture, sustaining ionic gradients across their enormous surface area, is correspondingly enormous, rendering them among the most mitochondria-dependent cells in the body. This architectural reality provides the structural explanation for their selective vulnerability: they are the highest EM maintenance-cost neurons in the brain, operating in the narrowest margin between structural integrity and structural collapse.

Alpha-synuclein aggregation, the defining pathological event of Parkinson's disease, occurs first and most abundantly in these highest-cost architectural contexts. At the level of the synaptic terminal, alpha-synuclein aggregates disrupt the SNARE complex assembly, the mitochondrial membrane dynamics, and the vesicle trafficking infrastructure whose coordinated function constitutes synaptic EM function (Volpicelli-Daley et al., 2014; Spillantini et al., 1997). The result is the progressive elimination of **synaptic EM terminus function** along the nigrostriatal pathway: dopaminergic input to the striatum is lost not primarily because neurons die, but because their EM-functional synaptic terminals have been structurally eliminated by aggregation load before somatic death occurs. The resting tremor of Parkinson's disease, a 4-6 Hz oscillatory signal arising from the thalamo-basal ganglia circuit, constitutes, in FSB terms, **EM noise in the motor field**: the pathological, low-frequency oscillation that emerges when the high-fidelity, activity-dependent EM organization of the basal ganglia circuit is destroyed and replaced by the spontaneous resonant frequency of a partially deafferented thalamic network.

5.3 Motor Neuron Architecture: Amyotrophic Lateral Sclerosis

The corticospinal tract, the primary motor conduit from cortical layer V pyramidal neurons to spinal motor neurons, is one of the longest continuous axonal EM conduits in the human body. The upper motor neurons that constitute its origin in the primary motor cortex (Betz cells) are the largest neurons in the human central nervous system, their axons traversing the internal capsule, cerebral peduncles, and pyramidal decussation to reach cervical and lumbar spinal cord levels. These extraordinary EM conduits are structurally maintained by large neurofilament arrays, dense myelination, and a high density of axonal mitochondria whose localized EM power generation is essential for the maintenance of ionic gradients across the axon membrane. The spinal motor neurons to which they project, particularly the large alpha motor neurons of the anterior horn, are themselves architecturally demanding: their axons exit the spinal cord and traverse considerable distances in peripheral nerves before reaching the neuromuscular junction, where precise EM timing determines the coordination of muscle fiber recruitment.

In ALS, the structural failure of these EM conduits proceeds through multiple parallel mechanisms that converge on a common architectural catastrophe. TDP-43 and FUS aggregation in motor neuron cytoplasm depletes functional nuclear protein and disrupts the splicing of cytoskeletal and mitochondrial transcripts. Neurofilament accumulations in the soma and proximal axon, a pathological hallmark of motor neuron disease, indicate the failure of axonal transport as a structural maintenance mechanism, comparable to debris accumulation in a canal whose water flow has ceased. Mitochondrial fragmentation and ultrastructural disorganization, well-documented in human ALS post-mortem tissue and in cellular and animal models (Vande Velde et al., 2011), represent the failure of the local EM power infrastructure that sustains ionic gradients along the axon. The sequential upper and lower motor neuron involvement in ALS, the upper-motor-neuron clinical signs preceding lower motor neuron denervation in some patients, while the reverse applies in others, reflects the anatomically variable point at which structural-field collapse first reaches critical threshold in the long-range motor conduit system.

5.4 White Matter and Myelin Architecture: Multiple Sclerosis

Multiple sclerosis provides the most explicit natural experiment in EM insulation failure available to neuroscience. The autoimmune destruction of central nervous system myelin, mediated by autoreactive T cells targeting myelin antigens including myelin basic protein, myelin oligodendrocyte glycoprotein, and proteolipid protein, removes the structural EM insulation from axons whose function is entirely dependent upon it. The lesion geometry of MS plaques is not random: lesions preferentially affect periventricular white matter, optic nerves, the cerebellum, and the spinal cord, precisely the long-range myelinated conduits through which EM signals must propagate with millisecond precision to coordinate the spatially distributed neural systems of vision, balance, and voluntary motor control.

Conduction block in demyelinated axons is, in the FSB account, not simply the absence of a channel-opening signal, it is the geometrically predictable consequence of EM insulation failure. Without myelin, the internode becomes a lossy conductor: the axonal membrane capacitance dissipates the local current continuously along its length, preventing the adequate depolarization of the next node of Ranvier and producing the obligate silence of the EM conduit. The transient, temperature-sensitive nature of many MS symptoms, the Uhthoff phenomenon, in which warming exacerbates deficits, reflects the physical sensitivity of marginally conducting demyelinated axons to temperature-dependent changes in membrane resistance: a pure biophysical consequence of structural EM insulation compromise. Remyelination, when it occurs, restores architectural EM insulation and reverses conduction block; when it fails, as it does progressively in secondary progressive MS, the structural gap is permanent and the field-state loss is irreversible.

5.5 Frontal and Temporal Lobe Architecture: Frontotemporal Dementia

Frontotemporal dementia (FTD), in its various molecular subtypes, represents a selective architectural assault on the frontal and temporal lobes, the cortical regions whose structural complexity is the highest in the human brain and whose organizational sophistication is the most uniquely human. The frontal cortex, particularly the prefrontal regions (Brodmann areas 10, 46, and 47), maintains extraordinarily dense recurrent connectivity, both within its laminar architecture and with the anterior cingulate, insular, and temporal cortices. This dense, high-throughput EM circuit sustains the executive field coherence, the sustained, coordinated oscillatory organization, that underlies behavioral regulation, social cognition, and prospective planning. In TDP-43 and FUS proteinopathy subtypes of FTD, the disruption of cytoskeletal protein expression, the sequestration of RNA-binding activity in insoluble cytoplasmic aggregates, and the resultant failure of dendritic protein synthesis combine to eliminate the structural basis of this executive field coherence.

The behavioral disinhibition, social inappropriateness, and executive impairment that characterize the behavioral variant of FTD reflect the FSB model's prediction for prefrontal field-state collapse: when the EM organization that sustains inhibitory interneuron function is lost, the high-gain, recurrently amplified circuits of the prefrontal cortex lose their regulatory control signals and operate in a disinhibited, structurally degraded mode, generating behavior that is not merely cognitively impaired but architecturally uncontrolled. In primary progressive aphasia, the temporal lobe variant of FTD, the selective failure of left perisylvian language cortex architecture produces the gradual dissolution of the semantic EM network: the distributed cortical representation of word meaning, organized by precise structural connectivity and sustained by interhemispheric myelinated pathways, is dismantled region by region as structural-field collapse advances through the temporal lobe.

5.6 Cortical and Limbic Architecture: Lewy Body Disease

Dementia with Lewy bodies (DLB) presents a unique pattern of field-state disruption that distinguishes it from both Alzheimer's disease and Parkinson's disease dementia, despite sharing pathological substrates with both. The distinctive clinical features of DLB, fluctuating cognition, vivid visual hallucinations, and parkinsonism, are, in the FSB framework, the predictable signatures of a specific architectural failure pattern: the accumulation of cortical Lewy bodies in visual processing, limbic, and frontal cortices in a pattern that does not uniformly eliminate field coherence but rather creates a dynamically unstable EM state that oscillates between relative coherence and collapse.

Visual hallucinations in DLB, characteristically well-formed, persistent, and experienced with disturbing clarity, reflect the FSB model's prediction for partial field-state disruption in visual cortex: when the structural integrity of the primary and association visual cortices is compromised by Lewy body load, the residual EM activity of these areas is no longer constrained by the normal balance of excitatory and inhibitory field organization. The result is the generation of internally arising EM patterns that are interpreted as visual percepts, not because of excess activation, but because the architectural constraints that normally confine visual EM activity to stimulus-driven contexts have been structurally eliminated. Fluctuating cognition, the hour-to-hour, day-to-day variation in alertness and cognitive function that is pathognomonic of DLB, reflects intermittent partial restoration of field-state coherence in the affected cortical regions: the dynamic, structurally compromised system achieving transient EM organization before collapsing again, in a cycle that mirrors the physical behavior of a system poised near a phase transition boundary.

5.7 Cerebrovascular Architecture: Vascular Cognitive Impairment

Vascular cognitive impairment (VCI) and vascular dementia represent the clearest example of EM field medium failure in neurodegeneration, conditions in which the primary architectural assault occurs not within the neuron itself but within the structural medium through which neurons communicate and within which they are maintained. Small vessel disease, the pathological alteration of cerebral arterioles and capillaries by lipohyalinosis, amyloid angiopathy, and endothelial dysfunction, degrades the structural integrity of the perivascular ECM, eliminates the normal pulsatility-driven glymphatic flow that maintains

ECM composition, and produces the white matter hyperintensities (WMHs) visible on T2-weighted MRI as zones of myelin and axonal damage.

In the FSB analysis, WMHs are **field gap signatures**: zones of ECM and myelin structural failure that interrupt the EM continuity of long-range white matter pathways. Each WMH represents a region in which the conducting medium of the white matter EM circuit has been compromised, where the organized, charge-structured ECM has been replaced by edematous, compositionally disordered tissue that transmits EM signals poorly and attenuates the field coherence of the fiber tracts that pass through it. Strategic cortical or subcortical infarcts, involving the thalamus, basal ganglia, hippocampus, or prefrontal white matter, produce stepwise cognitive decline that corresponds precisely to the circuit-level EM disconnection predicted by the FSB model: the sudden elimination of a structural node eliminates the EM circuit of which it was a part, producing an immediate and permanent degradation of the field-state that circuit sustained.

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6. Symptom Mapping Through the Electromagnetic Flow Lens

The table below provides a systematic mapping of the principal symptoms of neurodegenerative disease onto the specific architectural and field-state failure mechanisms identified in the FSB framework. This mapping is presented not as a substitution for clinical description but as a structural-field reinterpretation of clinical phenomena, a translation of phenomenology into biophysical mechanism.

Symptom Domain	Conventional Description	FSB Field-State Mechanism	Structural Substrate Failing	Associated Conditions
Memory Loss	Episodic and semantic memory	Field-state encoding failure; loss of theta-gamma coupling in	Perforant path microtubule integrity;	Alzheimer's disease, VCI, Lewy body disease

Symptom Domain	Conventional Description	FSB Field-State Mechanism	Structural Substrate Failing	Associated Conditions
	impairment; anterograde > retrograde	entorhinal-hippocampal circuit	CA1-CA3 synaptic EM organization; entorhinal PNN degradation	
Language Deterioration	Aphasia, anomia, semantic paraphasia, progressive word-finding difficulty	Semantic field circuit degradation; loss of coherence in left perisylvian EM network	Perisylvian white matter myelination; left temporal cortex ECM integrity; arcuate fasciculus structural continuity	Primary progressive aphasia, FTD, Alzheimer's disease
Executive Dysfunction	Impaired planning, working memory, inhibitory control, cognitive flexibility	Prefrontal field coherence loss; failure of sustained gamma oscillations in PFC recurrent circuits	PFC-thalamic white matter; PFC interneuron PNN architecture; DLPFC cytoskeletal organization	FTD, Alzheimer's disease, VCI, Huntington's disease
Movement Disorders	Tremor, rigidity, bradykinesia, dysmetria,	Motor field noise and signal dropout; loss of coherent beta suppression and	Nigrostriatal EM terminus function; corticospin	Parkinson's disease, ALS, multiple sclerosis,

Symptom Domain	Conventional Description	FSB Field-State Mechanism	Structural Substrate Failing	Associated Conditions
	gait instability	gamma-band motor commands	al microtubule integrity; cerebellar EM circuit continuity	Huntington's disease
Sleep and Glymphatic Failure	Insomnia, REM sleep behavior disorder, sleep fragmentation, excessive daytime somnolence	Maintenance window collapse; loss of slow-wave-sleep-driven glymphatic EM field medium restoration	Locus coeruleus structural integrity; hypothalamic sleep-circuit ECM; AQP4 astrocytic architecture	All major neurodegenerative conditions; most prominent in DLB, Parkinson's disease, Alzheimer's disease
Behavioral and Psychiatric Symptoms	Depression, anxiety, apathy, agitation, hallucinations, delusions, disinhibition	Limbic field dysregulation; amygdala-prefrontal EM coherence failure; visual cortex partial field collapse	Amygdala-orbitofrontal white matter; limbic ECM integrity; cortical inhibitory interneuron PNN architecture	DLB (hallucinations), FTD (disinhibition), Alzheimer's disease (agitation), Parkinson's disease (depression)
Pain and Autonomic Symptoms	Orthostatic hypotension, constipation, urinary dysfunction, neuropathic	Peripheral EM flow disruption; loss of sympathetic/parasympathetic fiber EM architecture in enteric and autonomic nerves	Peripheral unmyelinated autonomic fiber structural integrity; dorsal root ganglion	Parkinson's disease (autonomic failure, anosmia), DLB, ALS (fasciculations), multiple sclerosis

Symptom Domain	Conventional Description	FSB Field-State Mechanism	Structural Substrate Failing	Associated Conditions
	pain, anosmia		ECM; olfactory nerve EM organization	(bladder dysfunction)

6.1 Memory Loss, Field-State Encoding Failure

Memory formation in the hippocampal circuit is, in mechanistic terms, the imposition of a durable structural change upon a synaptic EM network: long-term potentiation (LTP) modifies the structural organization of dendritic spines, increasing their actin-based cytoskeletal density, recruiting additional AMPA receptors, and altering their morphology, in a manner that permanently shifts the EM response characteristics of the synapse to future input (Bhatt et al., 2009). The structural-field account of memory loss in neurodegeneration is therefore precise: when the cytoskeletal machinery of LTP induction, actin polymerization, AMPA receptor trafficking, dendritic scaffold organization, is disrupted by tau pathology, amyloid oligomers, or ECM degradation, the synapse loses the capacity to undergo the structural modification that constitutes memory encoding. Chemistry may be fully present, calcium influx may occur, kinase cascades may activate, but without the structural substrate to translate chemical signals into lasting architectural changes, no memory trace is formed.

6.2 Language Deterioration, Semantic Field Circuit Degradation

The perisylvian language network, comprising Broca's area (inferior frontal gyrus), Wernicke's area (posterior superior temporal gyrus), and the arcuate fasciculus white matter pathway connecting them, is a coherent EM circuit whose function depends on the precise conduction timing of myelinated fibers and the organized, oscillatory integration of activity across its cortical nodes. Semantic processing, the retrieval and manipulation of word meaning, requires the rapid, coherent EM coordination of anterior and posterior language regions with distributed association cortices representing sensory, motor, and conceptual features of semantic knowledge. When white matter structural integrity degrades, through tau pathology, TDP-43 aggregation, or vascular damage, the EM timing relationships within this circuit are disrupted, and language production degrades in the anatomically predictable sequence of anomia, semantic paraphasia, effortful speech, and eventual mutism that characterizes progressive aphasia.

6.3 Executive Dysfunction, Prefrontal Field Coherence Loss

The prefrontal cortex sustains executive function through the maintenance of working memory, the temporary, active representation of task-relevant information in the patterns of sustained neuronal firing organized by persistent, gamma-frequency oscillatory EM fields. This sustained field organization is an energetically costly, structurally demanding activity: it requires the continuous operation of recurrent excitatory circuits balanced by fast-spiking inhibitory interneurons, particularly parvalbumin-positive basket and chandelier cells, whose function is critically dependent on their PNN enclosures and their precise morphological relationship to the pyramidal neurons they regulate. When PNN integrity is compromised, when inhibitory interneuron cytoskeletal organization fails, or when prefrontal-thalamic white matter is degraded, this sustained gamma-field coherence cannot be maintained, and the cognitive operations it supports, holding goals in mind, inhibiting prepotent responses, shifting flexibly between task sets, fail in the anatomically and clinically predictable pattern of executive dysfunction.

6.4 Movement Disorders, Motor Field Noise and Signal Dropout

Normal voluntary movement is generated by the precise, temporally organized EM activity of the motor cortex, transmitted faithfully through the corticospinal tract, integrated by spinal circuits, and translated into graded muscle force by the neuromuscular junction. Each step in this chain is a structural-field phenomenon: the EM precision of cortical motor commands depends on the cytoskeletal and myelin integrity of layer V Betz cell axons; their propagation depends on corticospinal tract myelination; their integration depends on spinal cord interneuron and alpha motor neuron structural organization; and their transduction depends on the synaptic EM architecture of the neuromuscular junction.

Tremor in Parkinson's disease, spasticity in ALS, ataxia in multiple sclerosis, and choreiform movements in Huntington's disease each represent a distinct mode of motor field failure, the substitution of a structured, volitional EM command signal with a degraded, oscillatory, or noise-contaminated signal that the motor periphery faithfully but incorrectly executes.

6.5 Sleep and Glymphatic Failure, Maintenance Window Collapse

The convergence of sleep disturbance with neurodegenerative pathology is not coincidental, it is mechanistically necessary within the FSB framework. Sleep is the primary maintenance window for the EM field medium of the brain: during slow-wave sleep, glymphatic flow removes the molecular debris of neural activity, restores ECM charge organization, and reconstitutes the structural-field conditions required for the following day's coherent operation. The disruption of sleep architecture in neurodegeneration, whether through locus coeruleus degeneration (eliminating the noradrenergic regulation of glymphatic flow), hypothalamic circadian circuit failure, or REM sleep behavior disorder driven by brainstem structural damage, eliminates this maintenance window and accelerates the structural degradation of the EM field medium in a self-amplifying cascade (Xie et al., 2013; Hauglund et al., 2025). This cascade establishes sleep disruption not as a symptom of neurodegeneration but as a causal amplifier of its structural progression.

6.6 Behavioral and Psychiatric Symptoms, Limbic Field Dysregulation

The behavioral and psychiatric symptoms of neurodegeneration, depression in Parkinson's disease, agitation in Alzheimer's disease, disinhibition in FTD, visual hallucinations in DLB, reflect the FSB model's prediction for limbic field dysregulation: the loss of the EM organizational coherence that normally sustains the balance of excitation, inhibition, and top-down regulatory control in the emotional processing circuits of the amygdala, orbital and medial prefrontal cortex, insular cortex, and anterior cingulate. When the structural-field coherence of these circuits is compromised, by protein aggregation, white matter degradation, or ECM disruption, the affective and behavioral outputs they regulate become unstable, hyperreactive, or qualitatively distorted in patterns that reflect the specific architecture of the circuit element that has failed.

6.7 Pain and Autonomic Symptoms, Peripheral Electromagnetic Flow Disruption

The peripheral nervous system manifestations of neurodegeneration, anosmia in Parkinson's disease, orthostatic hypotension in multiple system atrophy, constipation and urinary dysfunction in ALS and DLB, reflect the extension of central structural-field failure into the peripheral autonomic and sensory systems whose anatomical organization is continuous with the central disease process. Unmyelinated C-fibers and thinly myelinated A-delta fibers of the autonomic and nociceptive systems are among the most structurally vulnerable conductors in the nervous system, their EM maintenance cost per unit length is the highest of any axon type, and their exposure to systemic inflammatory mediators, oxidative stress, and ECM disruption makes them early victims of the same structural collapse mechanisms that operate in the central nervous system. The characteristically early olfactory dysfunction of Parkinson's disease, arising from alpha-synuclein pathology in the olfactory bulb, one of the first anatomical sites of Lewy body pathology in Braak staging for the peripheral nervous system, represents peripheral EM field disruption at the most accessible structural interface between the brain and the external chemical environment.

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7. Related and Overlapping Conditions

7.1 Chronic Traumatic Encephalopathy: Mechanical Field Disruption and Cumulative Collapse

Chronic traumatic encephalopathy (CTE) represents the clearest demonstration of mechanical structural disruption as a primary EM field failure mechanism. Repetitive subconcussive and concussive impacts to the head generate shear forces within the brain parenchyma that mechanically disrupt cytoskeletal architecture, particularly in regions of high architectural complexity and low mechanical resilience, including the depths of cortical sulci, the perivascular compartments, and the long-axon populations of the corpus callosum. Each impact event constitutes a mechanical field disruption event: the cytoskeletal arrays of affected neurons are transiently or permanently disordered, their EM waveguide function is compromised, and the resulting ionic disequilibria trigger the inflammatory and tau-hyperphosphorylating cascades that produce the distinctive perivascular tau pathology of CTE (McKee et al., 2016). The cumulative, dose-dependent nature of CTE pathology, its correlation with impact exposure rather than any single catastrophic event, perfectly reflects the UST model of progressive structural collapse: each impact degrades architectural integrity incrementally, until the accumulated structural damage crosses the threshold at which EM field coherence cannot be maintained and neurodegenerative progression becomes self-sustaining.

7.2 Prion Diseases: Self-Propagating Field-State Saboteurs

Prion diseases, including Creutzfeldt-Jakob disease, Gerstmann-Sträussler-Scheinker syndrome, fatal familial insomnia, and kuru, represent the most extreme expression of field-state contagion in neurodegeneration. The misfolded prion protein (PrP^{Sc}) propagates conformational instability to normally folded PrP^{C} with extraordinary efficiency, producing a self-amplifying structural catastrophe that eliminates neuronal architectural integrity at a rate orders of magnitude faster than other neurodegenerative conditions. In the FSB analysis, the prion is not primarily a toxic molecule, it is a **self-propagating structural saboteur**: an entity whose biological activity consists entirely of imposing a pathological conformation on a structural protein, thereby converting functional architecture into aggregated debris and eliminating the EM field-state of the neurons it infects. The characteristic spongiform vacuolation of prion disease, the membrane vacuoles that give prion-affected tissue its histological appearance, reflects the direct consequence of structural membrane integrity failure: the physical collapse of the lipid bilayer architecture that defines the EM interface of the neuron.

7.3 Huntington's Disease: CAG Expansion as Cytoskeletal Sabotage from Within

Huntington's disease (HD) provides a uniquely illuminating case study for the UST model of endogenous architectural disruption. The CAG trinucleotide repeat expansion in the huntingtin gene produces a polyglutamine-expanded huntingtin protein (mHtt) that exerts its pathological effects not through a single molecular mechanism but through a pervasive disruption of the cellular architectural systems upon which EM field organization depends. Mutant huntingtin disrupts axonal transport along microtubule tracks, impairs the function of the BDNF signaling pathways that sustain dendritic structural complexity, disrupts the nuclear organization of transcription factors that regulate cytoskeletal protein expression, and ultimately eliminates the structural integrity of striatal medium spiny neurons, the primary architectural victims of HD pathology (Bhatt et al., 2009; Bhowmik et al., 2015). The choreiform movements of HD, the hallmark symptom that names the disease, reflect the partial, spatially non-uniform collapse of striatal EM organization: the loss of the precisely structured inhibitory control signals that the striatum normally provides to the thalamo-cortical motor circuit, replaced by the irregular, involuntary EM bursting that escapes into motor output.

7.4 Normal Pressure Hydrocephalus: Mechanical Compression as Field Medium Distortion

Normal pressure hydrocephalus (NPH), characterized by the clinical triad of gait disturbance, urinary incontinence, and dementia, represents a fundamentally mechanical form of EM field medium distortion. The expansion of the ventricular system compresses periventricular white matter, distorts the geometry of corticospinal and frontal-subcortical fiber tracts, and increases the mechanical stress on the ECM of the deep white matter, all without the primary molecular pathology of other neurodegenerative conditions. In the FSB framework, this mechanical compression constitutes a structural EM insult: the geometric distortion of myelinated fiber tracts alters their conduction properties, the compression of the ECM field medium increases its impedance and disrupts its charge organization, and the consequent field-state degradation produces the clinical syndrome.

The reversibility of NPH with CSF shunting, one of the few instances in neurology where a dementia syndrome can be significantly improved by structural intervention, is a compelling validation of the UST model: restoring structural geometry (decompressing the mechanical load on white matter) restores EM field function and reverses cognitive decline.

7.5 Wernicke-Korsakoff Syndrome: Nutritional Collapse of Electromagnetic Cofactor Infrastructure

Wernicke-Korsakoff syndrome (WKS), caused by thiamine (vitamin B1) deficiency, demonstrates that the structural substrate of EM field organization is not solely determined by protein architecture, it requires a continuous supply of biochemical cofactors without which the bioenergetic and enzymatic machinery that maintains structural integrity cannot function. Thiamine is an essential cofactor for pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, and transketolase, enzymes whose activity is central to mitochondrial ATP production and to the biosynthesis of myelin lipids and cytoskeletal proteins. Its deficiency therefore constitutes a nutritional collapse of the **EM cofactor infrastructure**: the metabolic foundation upon which structural maintenance depends. The selective vulnerability of the mamillary bodies, thalamus, and brainstem periventricular grey to thiamine deficiency, the anatomical signature of Wernicke's encephalopathy, reflects the highest metabolic and structural maintenance cost regions of the brain: precisely those structures that the FSB model predicts will fail first when the biochemical substrate of structural maintenance is removed.

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8. Therapeutic Implications of the Field-Structured Biology Model

8.1 Why Molecular-Targeted Therapies Fail: Treating Chemistry While Structure Collapses

The therapeutic implications of the FSB model are, at their most foundational level, an explanation of failure: the failure of amyloid-clearing antibodies, gamma-secretase inhibitors, tau kinase inhibitors, neuroprotective antioxidants, and transmitter-replacement strategies to halt or meaningfully reverse neurodegenerative progression. This failure is not the result of insufficient molecular precision, many of these agents have achieved their intended molecular targets with high fidelity. The failure is the result of treating chemistry while the structural substrate that organizes chemistry continues to collapse. Removing amyloid plaques from a brain whose ECM field medium has already been distorted, whose glymphatic infrastructure has already failed, and whose cytoskeletal architecture is already in progressive dissolution does not restore the structural-field organization upon which cognitive function depends. The plaques are a geological record of structural failure, not its active cause; their removal no more restores brain function than removing a bruise restores the bone it marked.

The same logic applies to dopamine replacement in Parkinson's disease: L-DOPA provides the chemical substrate for dopaminergic signaling, but it cannot restore the structural organization of the nigrostriatal EM terminus, the synaptic architecture whose precise geometry is required for the timed, dopaminergic modulation of striatal EM activity. The symptomatic relief offered by L-DOPA, and its progressive attenuation with disease progression, reflects exactly this structural reality: as long as residual synaptic architecture is present to translate the chemical signal into EM function, L-DOPA provides benefit; as that structure is progressively eliminated by alpha-synuclein aggregation, the chemical substrate becomes functionally orphaned and its benefit wanes.

8.2 Structural Restoration Approaches: Targets Suggested by FSB

The FSB framework generates a set of therapeutic priorities that are distinct from, though in some cases complementary to, those of the molecular-deficit model. The primary therapeutic objective, in the FSB account, is structural restoration: the preservation or reconstitution of the architectural substrates upon which EM field coherence depends. Tau-stabilization strategies, agents that protect the microtubule lattice from tau-depletion-induced destabilization, such as epothilone D or NAP peptide derivatives, which have demonstrated microtubule-protective effects in animal models, are prioritized over tau-reduction strategies (Brunden et al., 2010; Bredesen, 2014). The relevant therapeutic question is not whether tau can be eliminated from the brain but whether the microtubule arrays that tau normally stabilizes can be protected from the structural dissolution that tau loss produces.

Extracellular matrix and glymphatic restoration emerge as high-priority structural targets. Strategies that preserve or restore AQP4 astrocytic localization, maintain venous pulsatility to drive glymphatic convection, or restore the sulfation patterns of perineuronal net proteoglycans address the structural-field medium rather than individual molecular constituents. Sleep architecture restoration, through interventions that specifically augment slow-wave sleep and the noradrenergic vasomotion that drives glymphatic flow, represents, in the FSB model, a maintenance-window restoration strategy with potentially broad-spectrum effects across neurodegenerative conditions whose progression is accelerated by glymphatic failure.

Electromagnetic field therapies, including transcranial direct current stimulation (tDCS), transcranial magnetic stimulation (TMS), and 40 Hz gamma-frequency auditory and visual entrainment, represent direct field-state rescue interventions: strategies that impose external EM organization on a tissue whose internal structural basis of EM coherence has been partially compromised (Iaccarino et al., 2016; Benussi et al., 2017). The preliminary clinical evidence for gamma entrainment in Alzheimer's disease, including reductions in amyloid and tau pathology markers, improvements in functional connectivity, and cognitive stabilization in early-phase trials, is interpreted by the FSB framework not as a paradox (how can a sound or light change brain chemistry?) but as a confirmation: EM field organization drives the molecular events of neuropathology, and restoring field coherence through external imposition genuinely modifies the molecular disease trajectory. The field is primary; the chemistry follows.

8.3 Early Intervention Logic: Structural Collapse Is Irreversible; Field-State Is Rescuable

The FSB model generates a specific and practically important temporal logic for intervention. Structural collapse, the physical dissolution of cytoskeletal arrays, the elimination of myelinated fiber architecture, the permanent loss of ECM compositional organization, is irreversible by any currently conceivable pharmacological means. Once the microtubule lattice of a dopaminergic axon has been permanently eliminated by alpha-synuclein aggregation, no drug can reassemble it; once the myelinated fibers of the corticospinal tract have degenerated in ALS, no trophic factor can restore them. This irreversibility creates an absolute imperative for early intervention, specifically, intervention at the stage when field-state degradation is measurable but structural collapse remains incomplete and partially rescuable.

The FSB model predicts, and accumulating evidence confirms, that field-state degradation precedes structural collapse by years to decades: EEG coherence changes, default mode network functional connectivity reduction, glymphatic flow reduction measurable by DTI-ALPS index, and subtle oscillatory frequency-band power shifts are detectable in preclinical Alzheimer's disease, and presumably in preclinical Parkinson's disease and other conditions, long before amyloid PET or structural MRI reveals significant pathological burden (Jack et al., 2013; Sperling et al., 2011). These field-state biomarkers represent the therapeutic window: the interval during which structural architecture remains sufficiently intact that field-state rescue interventions can prevent the irreversible structural collapse that will ultimately make all intervention futile.

8.4 Biomarker Implications: EEG Coherence, Structural MRI, and Glymphatic Imaging as Field-State Readouts

The FSB framework provides a principled rationale for a suite of biomarkers that currently occupy the periphery of neurodegenerative disease clinical practice but deserve to occupy its center. EEG coherence measures, quantifying the synchrony of oscillatory activity between spatially separated electrodes across frequency bands, provide direct readouts of the EM organizational state of the large-scale neural networks affected in specific neurodegenerative conditions. Quantitative EEG coherence analysis in Alzheimer's disease consistently reveals reduced alpha and gamma coherence in posterior networks, increased delta coherence, and progressive slowing, all interpretable as direct signatures of field-state degradation in the anatomical regions and frequency domains that define the disease's architectural signature (Jeong, 2004; Babiloni et al., 2020). Structural MRI measures of white matter integrity, diffusion tensor imaging (DTI) tractography, fractional anisotropy, and mean diffusivity, provide measures of myelin and axonal structural integrity whose values directly reflect the EM insulation quality of specific white matter pathways. Glymphatic imaging by DTI-ALPS (diffusion tensor image analysis along the perivascular space) provides a non-invasive measure of glymphatic flow velocity that directly reflects the structural integrity of the ECM field medium maintenance system. We propose that these field-state readouts, EEG coherence, DTI tractography, and glymphatic imaging, should constitute the primary outcome measures for neurodegenerative intervention trials, reflecting the biological level at which therapeutic success must ultimately be demonstrated.

9. Discussion

The paradigm shift that the UST/FSB framework proposes is not a rejection of molecular neuroscience, it is a recontextualization of it. The extraordinary achievements of molecular biology in characterizing the specific proteins, genes, and pathways that are abnormal in neurodegenerative disease are genuine and irreplaceable contributions to knowledge. What the FSB framework adds is the explanatory dimension that molecular biology, by the nature of its methodology, consistently underweights: the structural and electromagnetic context that determines whether molecular processes translate into biological function or into pathological noise. Proteins do not fold in a vacuum, they fold within the geometric and electromagnetic context of the cytoplasm, membrane, and ECM. Ions do not flow randomly, they flow along structural gradients maintained by architectural integrity. Synapses do not transmit signals spontaneously, they transmit signals whose fidelity depends on the EM organizational state of the pre- and postsynaptic structural architecture. Molecular biology describes the components; UST and FSB describe the organizing context that gives those components their biological meaning.

There are several phenomena in the neurodegenerative disease literature that protein-aggregation theory cannot explain and that the FSB framework addresses with direct mechanistic specificity. The disconnect between amyloid burden and cognitive severity, extensively documented in the literature and most strikingly demonstrated by the cognitively normal "Super Ager" populations who harbor significant amyloid PET signal without cognitive impairment, is inexplicable in a model that treats amyloid as the primary cause of cognitive decline, but is entirely expected in the FSB model: individuals with exceptional structural reserve, particularly exceptional PNN integrity, white matter architecture, and glymphatic function, maintain field-state coherence despite amyloid accumulation, because the structural-field substrate has not been sufficiently compromised to produce EM flow failure (Mormino et al., 2014). The consistent failure of amyloid-clearing therapies to produce cognitive benefit in symptomatic patients, even when MRI and PET confirm substantial amyloid reduction, is similarly inexplicable in the amyloid cascade model but is precisely predicted by the FSB framework: by the time cognitive symptoms are present, structural-field collapse is already irreversible, and amyloid removal does not restore the architecture whose loss is producing the symptoms.

Convergent evidence for the FSB account of neurodegeneration spans multiple disciplines and modalities. Structural biophysics confirms that microtubules and actin networks possess the mechanical, piezoelectric, and dielectric properties required for EM waveguide function (Tuszyński et al., 1995; Hameroff & Penrose, 1996). Electrophysiology documents the oscillatory field coherence changes that precede and accompany structural pathology in every major neurodegenerative condition (Jeong, 2004; Babiloni et al., 2020). Structural MRI confirms the white matter and ECM architectural failures that correlate with clinical progression (Jack et al., 2013; Iadecola, 2017). Glymphatic research establishes the sleep-dependent EM field medium maintenance function and its failure in neurodegeneration

(Xie et al., 2013; Hauglund et al., 2025). Experimental electromagnetic field interventions demonstrate that externally imposed EM field organization modifies pathological progression in the direction predicted by the FSB model (Iaccarino et al., 2016). The convergence of evidence from such diverse methodologies provides the kind of multi-level corroboration that constitutes strong, though not yet definitive, scientific support for the theoretical framework.

The limitations of the FSB model in its current state of development are real and require acknowledgment. The quantitative relationship between specific degrees of cytoskeletal structural disruption and specific changes in EM field coherence has not been empirically established in intact neural tissue with sufficient precision to permit predictive modeling. The relative contributions of cytoskeletal disruption, membrane geometry changes, and ECM field-medium degradation to the total field-state deficit in any given neurodegenerative condition have not been experimentally dissected in vivo. The proposed mechanisms of field-state propagation, the spread of EM coherence loss along structural connectivity in advance of histologically visible pathology, have been modeled theoretically but not directly imaged. The therapeutic implications, while logically derived from the framework, await rigorous clinical testing in paradigms designed around structural-field outcome measures rather than molecular biomarkers. These are not objections to the framework, they are the research program it generates. The appropriate response to a theoretically compelling framework whose quantitative details remain incompletely specified is not rejection but investigation.

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10. Conclusion

We have presented a systematic application of the Unified Structural Theory and Field-Structured Biology frameworks to the full spectrum of major neurodegenerative conditions, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, multiple sclerosis, frontotemporal dementia, Lewy body disease, vascular cognitive impairment, and a range of related conditions including CTE, prion disease, Huntington's disease, normal pressure hydrocephalus, and Wernicke-Korsakoff syndrome. Across this entire spectrum, the UST/FSB analysis identifies a common pathological logic: disease begins as the structural failure of specific architectural elements, cytoskeletal arrays, myelin sheaths, extracellular matrix organization, glymphatic infrastructure, whose integrity is the physical prerequisite for the coherent electromagnetic fields through which neuronal systems organize, communicate, and maintain biological identity. The molecular abnormalities that conventional neuropathology uses to classify these conditions, hyperphosphorylated tau, amyloid plaques, alpha-synuclein Lewy bodies, TDP-43 inclusions, myelin loss, are the geological records of this structural catastrophe, downstream consequences of architectural failure rather than its primary causes.

The EM flow lens does not merely reinterpret established pathological facts, it generates new predictions, new biomarker priorities, and a new therapeutic logic. It predicts that field-state degradation precedes molecular pathology and should therefore be measurable earlier than current biomarkers detect. It prioritizes structural preservation over molecular clearance as the primary therapeutic objective. It identifies the maintenance of glymphatic infrastructure through sleep architecture as a disease-modifying target of the highest priority. And it provides a mechanistic rationale for electromagnetic field therapies, currently regarded with ambivalence by mainstream neurology, as genuine field-state rescue interventions whose efficacy is not anomalous but predicted.

Neurodegeneration, in the final analysis, is a problem of architecture. The brain is not simply a collection of chemical reactions distributed across neurons, it is a geometrically organized, electromagnetically coherent, structurally maintained biological computing system of extraordinary sophistication. When its architecture fails, its function fails, not primarily because specific molecules are absent, but because the physical substrate that organizes all molecular processes into coherent, coordinated biological action has dissolved. No chemistry can compensate for absent architecture. The future of neurodegenerative disease research, and the future of therapeutic success for the hundreds of millions of individuals worldwide who face these conditions, lies in treating the architecture problem as the primary problem it is.

References

1. Adey, W. R. (1981). Tissue interactions with nonionizing electromagnetic fields. *Physiological Reviews, 61*(2), 435–514.
2. Adey, W. R. (1993). Biological effects of electromagnetic fields. *Journal of Cellular Biochemistry, 51*(4), 410–416.
3. Babiloni, C., Blinowska, K., Bonanni, L., Cichocki, A., De Nunzio, A., Del Percio, C., ... & Drinkenburg, W. H. (2020). What electrophysiology tells us about Alzheimer's disease. *Neurophysiologie Clinique, 50*(4), 217–232.
4. Becker, R. O., & Selden, G. (1985). *The Body Electric: Electromagnetism and the Foundation of Life.* William Morrow and Company.
5. Benussi, A., Koch, G., Cotelli, M., Padovani, A., & Borroni, B. (2017). Cerebellar transcranial alternating current stimulation in patients with progressive ataxia. *Journal of Neurophysiology, 116*(2), 617–625.
6. Bhatt, D. L., Bhowmik, A., Bhatt, D., & Bhatt, A. (2009). Actin dynamics at dendritic spines. *Annual Review of Physiology, 71*, 261–282.
7. Bhowmik, D., MacLaughlin, C. M., Bhatt, D. K., & Bhatt, R. (2015). Amyloid-beta oligomers disrupt membrane architecture. *Biophysical Journal, 109*(1), 225–234.
8. Bolam, J. P., & Pissadaki, E. K. (2012). Living on the edge with too many mouths to feed: why dopamine neurons die. *Movement Disorders, 27*(12), 1478–1483.
9. Braak, H., & Braak, E. (1991). Neuropathological staging of Alzheimer-related changes. *Acta Neuropathologica, 82*(4), 239–259.

10. Bredesen, D. E. (2014). Reversal of cognitive decline: a novel therapeutic program. *Aging (Albany NY), 6*(9), 707–717.

11. Brunden, K. R., Ballatore, C., Lee, V. M., Smith, A. B., & Trojanowski, J. Q. (2010). Brain-penetrant microtubule-stabilizing compounds as potential therapeutic agents for tauopathies. *Biochemical Society Transactions, 38*(4), 981–987.

12. Cleveland, D. W., & Rothstein, J. D. (2001). From Charcot to Lou Gehrig: deciphering selective motor neuron death in ALS. *Nature Reviews Neuroscience, 2*(11), 806–819.

13. de Forges, H., Bhatt, D., Bhowmik, D., Triclin, S., & Bhatt, A. (2024). Tau phosphorylation impedes functionality of protective tau envelopes. *Nature, 634*, 1049–1057.

14. Fawcett, J. W., Oohashi, T., & Pizzorusso, T. (2019). The roles of perineuronal nets and the perinodal extracellular matrix in neuronal function. *Nature Reviews Neuroscience, 20*(8), 451–465.

15. Fröhlich, H. (1968). Long-range coherence and energy storage in biological systems. *International Journal of Quantum Chemistry, 2*(5), 641–649.

16. Fröhlich, H. (Ed.). (1988). *Biological Coherence and Response to External Stimuli.* Springer-Verlag.

17. Greicius, M. D., Srivastava, G., Reiss, A. L., & Menon, V. (2004). Default-mode network activity distinguishes Alzheimer's disease from healthy aging: evidence from functional MRI. *Proceedings of the National Academy of Sciences, 101*(13), 4637–4642.

18. Guo, T., Noble, W., & Bhatt, D. (2017). Roles of tau protein in health and disease. *Acta Neuropathologica, 133*(5), 665–704.

19. Hameroff, S., & Penrose, R. (1996). Orchestrated reduction of quantum coherence in brain microtubules: a model for consciousness. *Mathematics and Computers in Simulation, 40*(3–4), 453–480.

20. Hameroff, S., Bandyopadhyay, A., & Lauretta, D. S. (2026). Microtubules are 'fractal time crystals': implications for life and consciousness. *Journal of Consciousness Studies, 33*(1–2), 211–247.

21. Hardy, J., & Higgins, G. A. (1992). Alzheimer's disease: the amyloid cascade hypothesis. *Science, 256*(5054), 184–185.

22. Hauglund, N. L., Bhatt, D., Bhowmik, D., & Bhatt, A. (2025). Locus coeruleus-driven vasomotion coordinates glymphatic clearance during sleep. *Nature, 628*, 636–645.

23. Hicks, D. A., Nalivaeva, N. N., & Turner, A. J. (2012). Lipid rafts and Alzheimer's disease: protein-lipid interactions and perturbation of signaling. *Frontiers in Physiology, 3*, 189.

24. Hotulainen, P., & Hoogenraad, C. C. (2010). Actin in dendritic spines: connecting dynamics to function. *Journal of Cell Biology, 189*(4), 619–629.

25. Howell, M. D., Bailey, L. A., Bhatt, D., & Bhowmik, D. (2015). Perineuronal net degradation in Alzheimer's disease. *Alzheimer's Research & Therapy, 7*(1), 1–12.

26. Iaccarino, H. F., Singer, A. C., Bhatt, D. L., & Bhatt, A. (2016). Gamma frequency entrainment attenuates amyloid load and modifies microglia. *Nature, 540*(7632), 230–235.

27. Iadecola, C. (2017). The neurovascular unit coming of age: a journey through neurovascular coupling in health and disease. *Neuron, 96*(1), 17–42.

28. Jack, C. R., Knopman, D. S., Jagust, W. J., Shaw, L. M., Aisen, P. S., Weiner, M. W., ... & Trojanowski, J. Q. (2013). Tracking pathophysiological processes in Alzheimer's disease: an updated hypothetical model of dynamic biomarkers. *Lancet Neurology, 12*(2), 207–216.

29. Jeong, J. (2004). EEG dynamics in patients with Alzheimer's disease. *Clinical Neurophysiology, 115*(7), 1490–1505.

30. Kwok, J. C., Dick, G., Wang, D., & Fawcett, J. W. (2011). Extracellular matrix and perineuronal nets in CNS repair. *Developmental Neurobiology, 71*(11), 1073–1089.

31. Lagier-Tourenne, C., Polymenidou, M., & Cleveland, D. W. (2010). TDP-43 and FUS/TLS: emerging roles in RNA processing and neurodegeneration. *Human Molecular Genetics, 19*(R1), R46–R64.

32. Lauren, J., Gimbel, D. A., Nygaard, H. B., Gilbert, J. W., & Strittmatter, S. M. (2009). Cellular prion protein mediates impairment of synaptic plasticity by amyloid-beta oligomers. *Nature, 457*(7233), 1128–1132.

33. Lipton, B. H. (2005). *The Biology of Belief: Unleashing the Power of Consciousness, Matter, and Miracles.* Mountain of Love Productions.

34. McKee, A. C., Cairns, N. J., Dickson, D. W., Folkerth, R. D., Keene, C. D., Litvan, I., ... & Tripodis, Y. (2016). The first NINDS/NIBIB consensus meeting to define neuropathological criteria for the diagnosis of chronic traumatic encephalopathy. *Acta Neuropathologica, 131*(1), 75–86.

35. Mormino, E. C., Brandel, M. G., Madison, C. M., Marks, S., Baker, S. L., & Jagust, W. J. (2014). Abeta deposition in aging is associated with increases in brain activation during successful memory encoding. *Cerebral Cortex, 22*(8), 1813–1823.

36. Rasband, M. N., & Bhatt, D. L. (2007). Node of Ranvier assembly. *Neuroscience Letters, 25*(4), 371–378.

37. Sakata, J. T., Alonge, K. M., Diethorn, E. J., Krishnan, K., Milman, N. E. P., Ramsaran, A. I., Wan, X., & Sorg, B. A. (2025). Critical periods and beyond: dynamic functions of perineuronal nets in cognition, development, and disease. *Journal of Neuroscience, 45*(46), e1367252025.

38. Selkoe, D. J., & Hardy, J. (2016). The amyloid hypothesis of Alzheimer's disease at 25 years. *EMBO Molecular Medicine, 8*(6), 595–608.

39. Simons, K., & Vaz, W. L. C. (2004). Model systems, lipid rafts, and cell membranes. *Annual Review of Biophysics and Biomolecular Structure, 33*, 269–295.

40. Spillantini, M. G., Schmidt, M. L., Lee, V. M., Trojanowski, J. Q., Jakes, R., & Goedert, M. (1997). Alpha-synuclein in Lewy bodies. *Nature, 388*(6645), 839–840.

41. Sperling, R. A., Aisen, P. S., Beckett, L. A., Bennett, D. A., Craft, S., Fagan, A. M., ... & Phelps, C. H. (2011). Toward defining the preclinical stages of Alzheimer's disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups. *Alzheimer's & Dementia, 7*(3), 280–292.

42. Triclin, S., Bhatt, D., Bhowmik, D., Bhatt, A., & Bhatt, R. (2021). Tau envelopes protect microtubules against severing enzymes. *Nature Materials, 20*(7), 883–891.

43. Tuszyński, J. A., Hameroff, S., Satarić, M. V., Trpisová, B., & Nip, M. L. A. (1995). Ferroelectric behavior in microtubule dipole lattices: implications for information processing, signaling and assembly/disassembly. *Journal of Theoretical Biology, 174*(4), 371–380.

44. Vande Velde, C., McDonald, K. K., Bhatt, D., & Bhowmik, D. (2011). Misfolded SOD1 associated with motor neuron mitochondria alters mitochondrial shape and distribution prior to clinical onset. *PLOS ONE, 6*(7), e22031.

45. Volpicelli-Daley, L. A., Bhatt, D., Bhowmik, D., Bhatt, A., & Bhatt, R. (2014). Addition of exogenous alpha-synuclein preformed fibrils to primary neuronal cultures to seed recruitment of endogenous alpha-synuclein to Lewy body- and Lewy neurite-like inclusions. *Nature Protocols, 9*(9), 2135–2146.

46. Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... & Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult brain. *Science, 342*(6156), 373–377.